

PHOTOACOUSTIC COMMUNICATION FROM AIR TO UNDERWATER BASED ON LOW-COST PASSIVE RELAYS

SEMINAR REPORT

**SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF DEGREE OF**

BACHELOR OF TECHNOLOGY

in

Electronics and Communication Engineering

of

APJ Abdul Kalam Technological University

by

SREEHARI M

Univ. Reg.No. VAS21EC097



(AN ISO 9001:2008 CERTIFIED INSTITUTION)

Department of Electronics and Communication

Vidya Academy of Science & Technology

Thalakkottukara, Thrissur - 680 501

(<http://www.vidyaacademy.ac.in>)

October 2024

Department of Electronics and Communication Vidya Academy of Science & Technology

Thalakkottukara, Thrissur - 680 501

(<http://www.vidyaacademy.ac.in>)



(AN ISO 9001:2008 CERTIFIED INSTITUTION)

Certificate

This is to certify that the seminar report titled **PHOTOACOUSTIC COMMUNICATION FROM AIR TO UNDERWATER BASED ON LOW-COST PASSIVE RELAYS** is a bonafide record of the work carried out by **SREEHARI M (Univ. Reg.No. VAS21EC097)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology in Electronics and Communication Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2024-2025.

Anju Vincent
(Seminar Guide)
Assistant Professor
Dept.of ECE
Vidya Academy of Science &
Technology, Thrissur

Resmi K Rajan
(Seminar Co-ordinator)
Assistant Professor
Dept.of ECE
Vidya Academy of Science &
Technology, Thrissur

Dr. S. Swapna Kumar
Professor and Head
Dept.of ECE
Vidya Academy of Science & Technology
Thrissur

Acknowledgement

I wish to record my indebtedness and thankfulness to all those who helped me prepare this report titled **PHOTOACOUSTIC COMMUNICATION FROM AIR TO UNDERWATER BASED ON LOW-COST PASSIVE RELAYS** and present it in a satisfactory way.

First and foremost I thank God Almighty for His providence and for being the guiding light throughout the seminar.

I would like to express my sincere gratitude to **Ms.Resmi K Rajan**, Assistant Professor of Electronics and Communication Engineering Department for the support and co-ordination.

I would like to thank my guide **Ms. Anju Vincent** , Assistant Professor of Electronics and Communication Engineering Dept. for providing critical inputs in the preparation of this report. I also thank all other faculty members in our department for their guidance.

I am thankful to **Dr. S Swapna Kumar**, Head of Electronics and Communication Engineering Department, and our Principal **Dr. Sunitha C**, for their sole co-operation.

Finally, I would like to extend my sincere gratitude to friends who have always been helpful, in preparing and presenting the report and in the discussion following the presentation.

SREEHARI M

Reg. No. VAS21EC097

Seventh Semester B.Tech (2021 Admissions)

Vidya Academy of Science & Technology

Thrissur - 680 501.

October 2024

Abstract

With the development of ocean research, communications from the air to underwater (e.g., aircraft to submarines) are especially important. However, the existing communication methods have their own limitations and wireless communication between underwater and the air has remained an unsolved problem. This paper proposes a photoacoustic communication based on low-cost passive relays, which is capable of not only minimizing the energy loss both in water and in the air-water interface, but also relieving the difficulty of target seeking. Thus, we can get a six order of magnitude reduction of laser energy (25 J vs. 27 mJ) using passive relays whose cost is less than USD 1.00. It is also demonstrated that our photoacoustic communication can work decently under complex water situations including waving and pollution. This creationary photoacoustic application may become a practical way to communicate from the air to underwater.

Contents

ACKNOWLEDGEMENT	i
ABSTRACT	ii
LIST OF FIGURES	iv
LIST OF SYMBOLS AND ABBREVIATIONS	v
1 INTRODUCTION	1
1.1 Overview	1
1.2 Literature Review	3
1.3 Work	8
1.4 Outline of the Report	9
2 METHODS AND EXPERIMENTAL SECTION	12
2.1 Introduction	12
2.2 Experimental Section	14
2.3 Photoacoustic Effect	15
2.4 Passive Relays	16
2.5 Photoacoustic Communication Working	16
3 RESULT	20
4 FUTURE SCOPE	24
5 CONCLUSION	27
BIBLIOGRAPHY	31

APPENDIX

32

List of Figures

1.1	Block Diagram of Photoacoustic Air-Underwater Wireless Communication	3
2.1	Air to Underwater Communication Working	17
3.1	Communication demonstration in complex water environment	21
3.2	Comparison of Underwater Acoustic Wave and Electromagnetic Wave Attenuation	22

List of Symbols and Abbreviations

USD	U.S Dollars
PA	Photo Acoustic
UVLC	Underwater Visible Light Communication
ASCII	American Standard Code for Information Interchange
AI	Artificial Intelligence
IoT	Internet of Things

Chapter 1

INTRODUCTION

1.1 Overview

In recent years, the demand for reliable communication between airborne platforms and underwater systems has significantly increased, driven by advancements in marine exploration, oceanography, and defense applications. As technologies evolve, the need for effective and efficient communication between aircraft and underwater vehicles, such as submarines and autonomous underwater vehicles (AUVs), has become paramount. However, traditional communication methods, including electromagnetic waves and acoustic signaling, face considerable challenges at the air-water interface. Issues such as energy loss, signal reflection, and refraction impede effective information exchange, creating a critical need for innovative solutions.

Electromagnetic waves, while capable of high-speed transmission, are heavily attenuated when crossing from air into water. The high absorption and scattering of these waves in water restrict their effective range, especially in deeper or murkier waters. Furthermore, when electromagnetic signals encounter the air-water boundary, they often reflect rather than transmit, compounding the difficulties in maintaining a reliable communication link. Acoustic signaling, on the other hand, has been traditionally used for underwater communication due to its ability to travel long distances with minimal loss. However, acoustic signals can be significantly affected by environmental factors such as temperature variations, salinity, and underwater noise, which can lead to degraded performance in dynamic marine environments.

To address these challenges, photoacoustic communication has emerged as a promis-

ing alternative. This innovative method leverages low-cost passive relays to convert electromagnetic signals from the air into ultrasonic waves that can travel efficiently through water. By utilizing the photoacoustic effect where light energy is transformed into sound waves through thermal expansion this technique presents an energy-efficient and cost-effective means of overcoming the communication barriers posed by the air-water interface.

The core principle of photoacoustic communication lies in the ability of light to induce localized heating in a medium, which results in the generation of sound waves. When a pulsed laser illuminates a light-absorbing material, the rapid thermal expansion generates a pressure wave, effectively creating an ultrasonic signal. This process is particularly advantageous in underwater environments, as photoacoustic waves experience less attenuation than electromagnetic waves, allowing for more reliable transmission of information over significant distances.

This seminar report explores the underlying principles of photoacoustic communication, detailing the design and function of low-cost passive relays that facilitate this process. These relays serve as critical components, converting the incoming electromagnetic signals into ultrasonic waves efficiently, thus bridging the gap between air and underwater communication. Their cost-effectiveness makes them accessible for various applications, from military operations to marine research.

Experimental results and simulations further demonstrate the potential of this method. Studies indicate that photoacoustic communication can effectively minimize energy loss and improve communication reliability, even in complex underwater conditions characterized by wave interference and pollution. The ability to maintain a stable connection under challenging circumstances marks a significant advancement over traditional methods.

In conclusion, the development of photoacoustic communication represents a vital step forward in addressing the longstanding challenges associated with air-to-underwater communication. By combining the principles of photonics and acoustics, this approach not only enhances the efficiency of information transfer but also opens up new avenues for applications in marine and defense sectors. As research continues and technology advances, photoacoustic communication could play a crucial role in facilitating seamless integration between airborne and underwater operations, paving the way for more effec-

tive marine exploration and strategic defense capabilities.

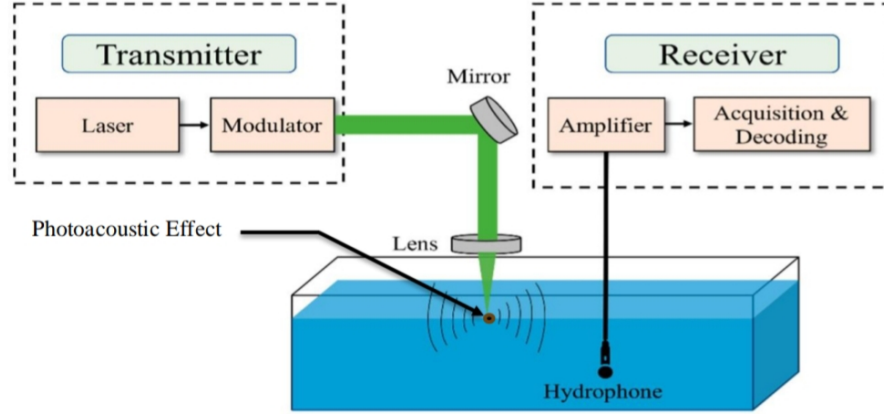


Figure 1.1: Block Diagram of Photoacoustic Air-Underwater Wireless Communication

Figure 1.1 illustrates a photoacoustic communication system for underwater data transmission. In the transmitter section, a laser generates a modulated light beam, which is directed and focused through a mirror and lens onto a specific point in the water. When the light reaches this point, it creates acoustic waves through the photoacoustic effect, where the absorption of light by water molecules causes localized heating and expansion, producing sound waves. These acoustic waves carry the encoded information and are detected by a hydrophone in the receiver section. The hydrophone captures the sound waves and sends the signal to an amplifier to increase its strength. Finally, the amplified signals are processed in the acquisition and decoding unit to retrieve the transmitted information. This system enables underwater communication by converting light signals into sound, making it suitable for environments where traditional electromagnetic waves cannot propagate effectively.

1.2 Literature Review

- Huazhen Chen, Yujiao Shi, and Da Xing, presented a paper on Photoacoustic Thermorelaxation Microscopy for Thermal Diffusivity Measurement, Thermal diffusivity is one of the main parameters to characterize the thermo-physical properties of

materials, and advances in its measurement technique will have significant impact on materials science and related applications. Here a photoacoustic (PA) thermorelaxation microscopy is proposed as a new noncontact method to measure the thermal diffusivity. By delivering co-focused heating/probing laser pulse pairs with tunable time delays, the sample's *in situ* thermal relaxation behavior after the heating pulse excitation can be photoacoustically monitored based on the temperature-dependent property of the Grueneisen parameter. We theoretically deduced the dependence of the obtained PA thermorelaxation time on the thermal diffusivity, and the results coincided well with simulations. The feasibility of this method was validated by various industrial and biological samples. This method provides a new strategy for high-resolution thermal diffusivity measurement with flexible measurement conditions, prefiguring great potential for material and biological applications.

- Fletcher Blackmon, Lee Estes, and Gilbert Fain, presented a paper on Linear Optoacoustic Underwater Communication, The linear mechanism for optical-to-acoustic energy conversion is explored for optoacoustic communication from an in-air platform or surface vessel to a submerged vessel such as a submarine or unmanned undersea vehicle. The communication range that can be achieved is addressed. A number of conventional signals used in underwater acoustic telemetry applications are shown to be capable of being generated experimentally through the linear optoacoustic regime conversion process. These results are in agreement with simulation based on current theoretical models. A number of practical issues concerning linear optoacoustic communication are addressed that lead to a formulation of a linear-regime optoacoustic communication scheme. The use of oblique laser beam incidence at the air–water interface to obtain considerable in-air range from the laser source to the in-water receiver is addressed. Also, the effect of oblique incidence on in-water range is examined. Next, the optimum and suboptimum linear optoacoustic sound-generation techniques for selecting the optical wavelength and signaling frequency for optimizing in-water range are addressed and discussed. Optoacoustic communication techniques employing M -ary frequency shift keying and multifrequency shift keying are then compared with regard to communication parameters such as bandwidth, data rate, range coverage, and number of lasers

employed.

- N. S. Gupta, presented a paper on Asymmetric Satellite-Underwater Visible Light Communication System for Oceanic Monitoring, In this paper, investigate the performance of the proposed oceanic monitoring system that connects the oceanic life with the terrestrial life. For continuous real-time monitoring and ubiquitous coverage, the communication system is aided with a satellite link. Multiple sensor nodes (SN) are deployed at different water levels that collect sensor data and transmit it to underwater vehicles (UV) using underwater visible light communication (UVLC). The UVLC system provides higher data rates at lower latency as compared to existing radio frequency (RF) and acoustic wave alternative for underwater communication (UWC). The UWC system comprises of horizontal haul (HH) and vertical haul (VH) UVLC links modelled using turbulence induced fading. The vertical haul links are modelled as the concatenation of successive non-mixing turbulent links to take into account the change of turbulence with the change of water level. The UVs and submarines communicate with the floating vessels (FVs) using vertical haul UVLC link. The UVs collect the data from the low power sensor nodes and offloads it to the FVs, which further beams it to the satellite on the RF carriers. The novel expressions of performance metrics such as outage probability and average bit error rate are derived. Further, the performance of the system is analysed for various system and channel parameters to prove the feasibility of the proposed communication system.
- S. Zong, presented a paper on Study of Opto-Acoustic Communication Between Air and Underwater Carrier, How to solve the communication problem to the underwater target has turned into one of the subjects that the militarists of all over the world commonly concern. Laser-induced acoustic signal is a new approach for underwater acoustic source, which has much virtue such as high intensity, short pulse and broad frequency. The paper studies the opto-acoustic communication method. The acoustic signal characteristic of laser-induced breakdown is studied and corresponding theory model is systemically analyzed. The opto-acoustic communication experimental measure investigation is formed with the high power laser, water tank and high frequency hydrophone. The characteristic of acoustic

signal is analyzed, such as intensity and frequency. This makes a stride for pursuing the feasibility of laser-acoustic underwater communication.

- F. Blackmon, L. Estes, and G. Fain, presented a paper on Linear Optoacoustic Underwater Communication, The linear mechanism for optical-to-acoustic energy conversion is explored for optoacoustic communication from an in-air platform or surface vessel to a submerged vessel such as a submarine or unmanned undersea vehicle. The communication range that can be achieved is addressed. A number of conventional signals used in underwater acoustic telemetry applications are shown to be capable of being generated experimentally through the linear optoacoustic regime conversion process. These results are in agreement with simulation based on current theoretical models. A number of practical issues concerning linear optoacoustic communication are addressed that lead to a formulation of a linear-regime optoacoustic communication scheme. The use of oblique laser beam incidence at the air–water interface to obtain considerable in-air range from the laser source to the in-water receiver is addressed. Also, the effect of oblique incidence on in-water range is examined. Next, the optimum and suboptimum linear optoacoustic sound-generation techniques for selecting the optical wavelength and signaling frequency for optimizing in-water range are addressed and discussed. Optoacoustic communication techniques employing M -ary frequency shift keying and multifrequency shift keying are then compared with regard to communication parameters such as bandwidth, data rate, range coverage, and number of lasers employed.
- Md Shafiqul Islam, Mohamed Younis, Munstasir Mahmud, Grey Carter, Fow-Sen Choa, presented a paper on A peak detection based OOK photoacoustic modulation scheme for air to underwater communication, The popularity of underwater applications has been growing due to the major technological advances in the last few decades. Yet, direct communication from an airborne node to an underwater node remains challenging due to the inability of any known signal which propagates well in both the air and water mediums. Photoacoustic energy transfer is a promising mechanism for enabling such cross-medium communication. Although the use of the photoacoustic mechanism is quite common in medical imaging, lit-

tle progress has been made on building the communication protocol stack. Most research studies have only focused on characterizing the channel and relating the laser and acoustic signals. Little attention has been given to developing suitable modulation and demodulation schemes. This paper fills such a technical gap and proposes a novel OOK modulation scheme using peak detection technique. We validate the effectiveness of our proposed scheme through simulation and lab experiments.

- Mahmud M, Islam MS, Ahmed A, Younis M, Choa FS, presented a paper on Cross-Medium Photoacoustic Communications: Challenges, and State of the Art, The current era is notably characterized by the major advances in communication technologies. The increased connectivity has been transformative in terrestrial, space, and undersea applications. Nonetheless, the water medium imposes unique constraints on the signals that can be pursued for establishing wireless links. While numerous studies have been dedicated to tackling the challenges for underwater communication, little attention has been paid to effectively interfacing the underwater networks to remote entities. Particularly it has been conventionally assumed that a surface node will be deployed to act as a relay using acoustic links for underwater nodes and radio links for air-based communication. Yet, such an assumption could be, in fact, a hindrance in practice. The paper discusses alternative means by allowing communication across the air–water interface. Specifically, the optoacoustic effect, also referred to as photoacoustic effect, is being exploited as a means for achieving connectivity between underwater and airborne nodes. The paper provides background, discusses technical challenges, and summarizes progress. Open research problems are also highlighted.
- Xiong Wang, Tao Qin, Yexian Qin, Russell S. Witte, Hao Xin, presented a paper on Microwave-Induced Thermoacoustic Communication, Wireless communications from an airborne device to an underwater receiver is challenging because the conductivity of water severely dampens the magnitude of electromagnetic waves in the spectrum commonly used for wireless applications. In this paper, microwave-induced thermoacoustic communications (TAC), a hybrid technique for wireless communications, is investigated and experimentally demonstrated to provide a po-

tential solution to this deficiency. TAC is based on the thermoacoustic effect and directly converts electromagnetic energy in air into acoustic energy in water. Experimental demonstration of TAC is reported by successful wireless information transmission from a microwave antenna in air to an acoustic transducer in water. Bench-top TAC experiments are combined with an analytical model to study the influences of pivotal design parameters on the performance of TAC. The experiments and theoretical analysis suggest that TAC might lead to a new paradigm of air-to-underwater wireless communications.

1.3 Work

The study focuses on the photoacoustic effect, where light energy is converted into ultrasound using passive relays that are cost-effective and highly efficient. The approach minimizes energy loss and overcomes challenges in complex water environments such as wave interference and pollution.

1. Photoacoustic Mechanism: The system utilizes a laser that emits coded pulses to passive relays, which generate ultrasonic signals that travel underwater. The relays are cost-effective, and the method reduces energy consumption significantly compared to existing technologies.

2. Efficiency in Complex Conditions: It outperforms pure light communication methods that struggle with water's refractive effects and pollution. The method ensures stable communication even in challenging environments with waves and pollutants.

3. Applications: This technology is valuable for military, marine research, and underwater exploration, especially in environments where traditional electromagnetic communication fails.

4. Experimental Setup: The system demonstrated efficient communication at lower energy costs using passive materials, and future improvements could enhance speed and distance further.

1.4 Outline of the Report

The first chapter, **Introduction**, sets the stage by addressing a key challenge in marine research and defense: establishing effective communication between airborne and underwater systems. This communication is essential for enhanced data gathering, environmental monitoring, and defense operations in marine environments. However, due to the contrasting properties of air and water, this task is met with considerable technical difficulties. Traditional methods primarily electromagnetic and acoustic systems suffer from significant energy loss at the air-water boundary, resulting in diminished signal strength and reduced efficiency. In Section 1.1, a summary of these current methods is provided. For example, while electromagnetic waves are efficient in air, they attenuate rapidly in water. Conversely, acoustic waves are suitable for underwater transmission but face barriers at the boundary. These limitations underscore the need for a novel approach to bridge this cross-medium communication gap.

Section 1.2 includes a **Literature Review** that explores previous studies tackling this issue. The review highlights research into optoacoustic systems, which convert light to sound, as well as microwave-based approaches aimed at facilitating communication across the air-water divide. Despite some promising results, these methods often encounter obstacles like excessive energy consumption, technical complexity, or high costs. Optoacoustic systems, though efficient in energy terms, require sophisticated equipment. Microwave-based techniques, while useful in shallow or clear waters, are less effective in deeper or turbid environments. These findings point to a demand for a more robust, energy-efficient solution to address communication challenges in varied aquatic settings.

Section 1.3, **Scope of the Work**, outlines the focus of this report: the development and testing of an innovative photoacoustic communication system using passive relays. This system leverages the photoacoustic effect, converting light energy into acoustic waves underwater to allow for more reliable and energy-efficient communication. The integration of passive relays, in particular, aims to reduce maintenance and operational costs while increasing the scalability of the system for extensive deployment.

The second chapter, **Methods and Experimental Section**, provides a technical overview of the experimental setup and methodology. Section 2.1, titled **Introduction**, explains the photoacoustic effect, which is used to convert electromagnetic waves in the air into

ultrasonic waves underwater. This effect is achieved when pulsed laser light contacts a material at the air-water boundary, causing rapid thermal expansion and producing ultrasonic waves. This method offers a unique advantage by reducing energy losses typically associated with traditional communication methods while enhancing cross-medium communication.

Section 2.2, **Experimental Setup**, provides a detailed account of the equipment and materials used in the experiments. The setup includes high-powered lasers that emit timed light pulses, ultrasonic detectors, amplifiers, and a series of passive relays. These passive relays must withstand repeated thermal expansions, ensuring stable operation and consistent signal quality. Calibration steps, such as laser alignment and relay placement, are also outlined to optimize measurement accuracy. By carefully structuring this setup, the experiment aims to demonstrate the effectiveness of the photoacoustic system in controlled conditions.

Section 2.3, **Photoacoustic Effect**, delves deeper into the mechanics of converting light energy to sound through thermal expansion. Unlike electromagnetic waves, which quickly diminish in water, photoacoustic waves maintain intensity, enabling longer transmission distances and consistent communication. This section highlights the efficiency of the photoacoustic effect compared to traditional methods and discusses the adaptability of this approach in complex water conditions, such as those with pollutants or wave interference.

Section 2.5, **Photoacoustic Communication Working**, visually presents the system's architecture. The block diagram illustrates how each component from the laser, which initiates the signal, to the passive relays that transmit it works together to facilitate communication across the air-water boundary. The diagram also shows the propagation of ultrasonic waves underwater, where detectors capture and decode the signals. This visual representation aids in understanding the system's flow and underscores the innovation behind this approach.

The third chapter, **Results**, presents the experimental findings. Section 3.1, **Communication Demonstration**, showcases the system's performance in various water environments, including clear, turbid, and wavy conditions. The results demonstrate that the photoacoustic system can consistently transmit signals even under challenging conditions, unlike traditional methods that struggle with interference from water waves or

contaminants. Supporting figures and data provide visual evidence of these outcomes, displaying the clarity and stability of the signal transmission.

In Section 3.2, **Signal Intensity Comparison**, the data illustrate the advantages of photoacoustic waves over traditional light-based signals in polluted or turbulent waters. Results reveal that the photoacoustic method retains higher signal intensity, minimizing energy loss across the air-water boundary. This improved efficiency not only enhances the system's potential range but also suggests its suitability for large-scale applications, providing reliable air-to-underwater communication in diverse marine settings.

Chapter four, **Future Scope**, explores further advancements for this technology. Suggested improvements include enhancing signal clarity, developing more resilient passive relay materials, and integrating the system with IoT networks for underwater data monitoring. This chapter also considers scalability, addressing how the system might be adapted for extensive ocean monitoring or military operations. Additionally, the potential for reduced costs through technological advancements is discussed, outlining a feasible path toward making photoacoustic communication more accessible and practical.

The fifth chapter, **Conclusion**, summarizes the research findings and their broader implications. The study concludes that the photoacoustic communication system provides a promising alternative to traditional methods by minimizing energy loss and boosting the reliability of air-to-underwater communication. The system's stability in complex water environments, including those with pollutants and waves, points to its significant potential for future applications in marine research and defense. By offering a robust, cost-effective, and scalable solution to the persistent challenge of cross-boundary communication, this technology paves the way for new advancements in underwater exploration and data collection.

Chapter 2

METHODS AND EXPERIMENTAL SECTION

2.1 Introduction

For many years, establishing effective wireless communication between air and underwater environments has been an unresolved challenge. As marine technology advances and the exploration of underwater realms becomes more critical, the need for reliable communication between these two domains has grown. In 2018, a research team from MIT made strides in this field by developing a one-way communication system that transmitted signals from underwater to the air using translational acoustic-radio frequency technology. However, the reverse sending signals from the air to underwater remains limited by current technologies, which face various practical constraints.

Direct communication between air and underwater is particularly difficult. Light-based communication systems work efficiently in both air and water environments, but when a laser beam reaches the air-water interface, it suffers from significant reflection and refraction, leading to energy losses and reduced accuracy, especially in the presence of waves. While blue and green light experience relatively low attenuation in pure water, this advantage diminishes in complex environments, such as those containing plankton, due to the high scattering of short-wavelength light. Other electromagnetic waves also encounter significant attenuation, causing substantial energy loss. Specifically, while microwaves and radio waves are ideal for air-based communication, they cannot effectively propagate through water.

Acoustic communication, particularly using ultrasonic waves, faces similar challenges. Although attenuation is minimal in both air and water, the large acoustic impedance difference at the air-water interface causes ultrasonic waves to lose most of their energy when crossing between the two mediums. To overcome these limitations, conventional air-to-underwater communication often relies on active relays. These devices, which require an external power source, actively convert signals (such as radio or light waves) from air into underwater signals (like ultrasonic or light waves), similar to a traditional radio receiver that re-broadcasts signals as sound waves below the water's surface. Active relays are typically installed on fixed offshore stations, buoys, or ships. However, they have several drawbacks. First, they are costly and require regular maintenance, leading to high operational expenses. Additionally, providing full coverage across the open ocean using active relays is challenging, which can result in communication failures. Furthermore, their relatively large size raises security concerns in certain applications. Given these limitations, there is a pressing need for passive relays that can facilitate air-to-underwater communication.

One promising approach to address these challenges is converting electromagnetic waves into ultrasonic waves at the air-water interface. This method leverages the fact that electromagnetic waves travel well in air, while ultrasonic waves can propagate efficiently underwater with minimal energy loss. Several mechanisms can be used to achieve this conversion, including thermal expansion, surface evaporation, and optical breakdown. In this context, the water at the air-water interface can serve as a potential passive relay. Various research teams have made notable progress in this area. For example, Wang et al. employed pulsed microwaves to generate ultrasonic waves through thermal expansion, utilizing the relatively high absorption coefficient of water in the microwave band. However, since microwaves tend to diverge, a high-power microwave source is required to deliver enough energy to the relay.

Similarly, other researchers have explored the use of high-power lasers in the visible and near-infrared spectrum to produce ultrasonic waves, due to the low absorption coefficients in these bands. However, these lasers require single pulse energies in the tens of joules range, which severely limits their practical application due to the high energy demand.

The photoacoustic effect, also known as the optoacoustic effect, offers a more effi-

cient means of converting light into ultrasound. In this process, light energy is absorbed by a material, which heats up and undergoes thermal expansion, generating a photoacoustic wave. When a pulsed laser illuminates an absorber under specific thermal and stress confinement conditions, a small temperature rise is sufficient to produce a photoacoustic wave. This effect has been widely used in biomedical imaging and various industrial applications. Additionally, using contrast agents with high optical absorption coefficients can further enhance the efficiency of this energy conversion, as the strength of the photoacoustic signal is proportional to the material's absorption coefficient. These characteristics make the photoacoustic effect a promising candidate for high-efficiency, air-to-underwater wireless communication.

In this paper, a novel communication system that utilizes low-cost, high-efficiency passive relays based on the photoacoustic effect. These passive relays can overcome the significant energy losses typically associated with electromagnetic waves in water and the high reflection that occurs when pure ultrasonic waves attempt to cross the air-water interface. By incorporating passive relays with high optical absorption coefficients, the system can significantly improve communication efficiency, achieving a six-order-of-magnitude reduction in laser energy. Specifically, this system requires only 27 millijoules of energy per pulse, compared to the 25 joules required by traditional methods. Moreover, the passive relays used in this system are inexpensive, costing less than USD 1.00 each.

Through simulations and experimental validation, it is demonstrated that this new photoacoustic-based approach has the potential to become a practical solution for wireless communication from air to underwater environments. By leveraging the advantages of passive relays and the efficiency of the photoacoustic effect, this method could pave the way for more reliable, cost-effective, and energy-efficient communication systems in marine technology and underwater exploration.

2.2 Experimental Section

The experimental section of the document describes the methods and setup for the photoacoustic air-to-underwater communication experiment.

- **Equipments Used:** A laser(custom-made, Beijing Laize, China) triggered by synchronization pulse of the function generator (AFG2021, Tektronix, U.S), Ultra-

sonic detector (5MHz:I5P6NF, Doppler, China; or 500KHz:DYW-500-E, Dayu Electric, China), Home-made amplifier with a magnification of about 40dB, Oscilloscope (MDO3054, Tektronics, U.S).

- **Signal Generation:** The laser, triggered by the synchronization pulse from a function generator, emits a 660nm wavelength beam with a 14ns pulse width at a 1 KHz repetition frequency. Upon striking an absorber, this laser generates ultrasonic waves through the photoacoustic effect. These waves, once received, undergo a series of processes: they are first amplified to enhance signal strength, then acquired by sensors, and finally decoded for interpretation, allowing the transfer of detailed information from the generated photoacoustic signals.
- **Experimental Setup:** The system employs a low-energy laser that operates at 27 mJ per pulse, combined with a passive relay, to enable efficient and reliable photoacoustic communication. This setup is designed to minimize energy consumption while maximizing communication range and signal clarity across the air-water boundary.

This experiment successfully demonstrates the viability of photoacoustic communication between air and underwater environments under complex conditions, including wave interference and water pollution.

2.3 Photoacoustic Effect

The photoacoustic effect, also known as the optoacoustic effect, is a highly efficient mechanism for converting light energy into ultrasonic energy at light absorbers via thermal expansion. When a light-absorbing material is illuminated with a pulsed laser, a small but rapid temperature increase occurs, generating a photoacoustic wave under conditions of thermal and stress confinement. This effect has broad applications in both biomedical fields and industrial settings, as photoacoustic waves have less attenuation than electromagnetic waves, allowing for deeper imaging. Photoacoustic waves travel more effectively in water than electromagnetic waves, enhancing their utility in aqueous environments.

2.4 Passive Relays

Passive relays are electrical components that enable the control of larger circuits using a smaller input signal, without needing external power for operation. When a low-power signal, such as voltage or current, is applied, the relay activates its contacts to either connect or disconnect, thereby controlling the flow of electricity in the main circuit. These relays typically consist of conductive contacts that open or close the circuit. In the case of electromagnetic relays, a coil generates a magnetic field that moves the contacts.

Common types of passive relays include electromagnetic relays, known for their reliability, and reed switches, which feature two ferromagnetic contacts sealed in a glass envelope that close in response to a magnetic field. These relays are widely used in control circuits for home appliances, industrial machinery, and automotive systems, allowing low-voltage signals to manage high-voltage circuits.

Additionally, passive relays are utilized in telecommunications and audio equipment for signal switching and in isolating different system components to protect sensitive parts from high voltages. While they offer benefits such as simplicity, low power consumption, and cost-effectiveness, passive relays may not switch as quickly as active components and can experience wear over time due to their mechanical nature.

Passive relays play a vital role in various electronic and electrical systems, providing an efficient method for controlling circuits with low-power signals.

2.5 Photoacoustic Communication Working

The photoacoustic air-to-underwater communication system represents an innovative method of transmitting data from air to underwater environments using laser pulses and the photoacoustic effect. The fundamental concept of this system involves a sender, typically situated in the air, such as on an aircraft, which emits a pulsed laser beam carrying encoded information. This pulsed laser beam is directed at a signal relay that acts as an intermediary between the air and water. The signal relay is a specially designed light-absorbing material, which is typically deployed by an underwater receiver, such as a submarine. Upon contact with the laser, the signal relay undergoes a process known as the photoacoustic effect, where the absorbed laser energy is converted into ultrasonic sig-

nals, referred to as photoacoustic waves. These ultrasonic waves then propagate through the water, carrying the transmitted data to the underwater receiver, which decodes the information.

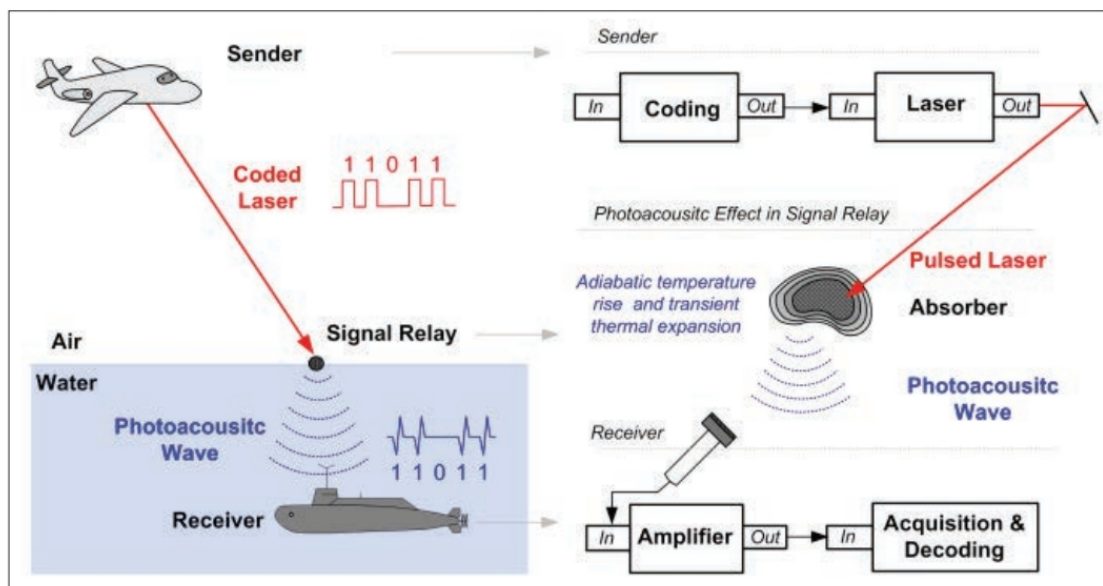


Figure 2.1: Air to Underwater Communication Working

The photoacoustic effect, central to this communication system, occurs when pulsed laser light strikes an absorber, causing a small, transient rise in temperature in the absorbing material. This temperature increase induces a rapid expansion and contraction of the material, which generates acoustic waves, or sound waves, in the form of ultrasonic pulses. These pulses travel through the water, where they are detected by the receiver and subsequently decoded back into the original data. The photoacoustic effect is highly effective for this kind of application because it allows the conversion of optical signals (light) in air into acoustic signals (sound) in water, which can travel more efficiently underwater.

In this communication system, a custom-made laser is used to produce the pulsed laser beam that initiates the photoacoustic effect. According to the experimental setup, the laser emits light at a wavelength of 660 nm (nanometers) and has a pulse width of 14 ns (nanoseconds), meaning the duration of each laser pulse is very short. This laser operates at a repetition frequency of 1 kHz, which indicates that it generates 1,000 pulses per second. The laser system serves as the information sender, and it is controlled by a function generator, which modulates the laser pulses to encode the desired information.

The laser used in the experiment has a single pulse energy of 27 mJ (millijoules), meaning that each pulse carries a small amount of energy sufficient to activate the photoacoustic effect without causing significant damage to the signal relay material.

The signal relay, or absorber, plays a critical role in this system as it acts as the medium that converts the optical energy from the laser into ultrasonic signals. When the pulsed laser beam hits the absorber, the energy is absorbed, causing the material to heat up. This rapid heating leads to thermal expansion, which, due to the transient nature of the temperature rise, is followed by rapid contraction. The alternating expansion and contraction generate the ultrasonic waves, which propagate through the water. These ultrasonic waves carry the encoded information, which can then be received and decoded by the underwater system.

The receiver in this system is designed to detect the ultrasonic waves generated by the signal relay. The experimental setup includes an ultrasonic detector that operates at a frequency of 5 MHz (megahertz), meaning it can detect ultrasonic signals with high precision. The detector is connected to a home-made amplifier that magnifies the detected signal by 40 dB (decibels), ensuring that even weak signals are sufficiently amplified for further processing. After amplification, the signal is acquired by an oscilloscope, which is used to visualize the signal and capture its waveform. This data is then sent to a computer, where the original information is decoded and recovered.

One of the most remarkable achievements of this system is its ability to reduce the energy required for transmitting signals by several orders of magnitude. The experimental results show that the system generates photoacoustic signals with voltages ranging from approximately 100 mV to 1 V. Despite the low energy of the laser pulses (27 mJ per pulse), the system is able to achieve the same level of signal generation that would typically require a much higher energy input (25 J). This significant reduction in energy requirement by a factor of six orders of magnitude demonstrates the efficiency of the photoacoustic communication system. Furthermore, the system uses low-cost passive relays, which cost less than USD 1.00, making it a highly cost-effective solution for long-distance underwater communication.

Another important feature of this system is the use of a low-power light source, which further enhances the efficiency of the communication system. By reducing the power of the laser, the system minimizes energy consumption, which is a critical consideration

for practical applications, particularly in remote or resource-constrained environments. The use of low-power lasers also reduces the risk of overheating or damaging the signal relay material. However, the system is not without its challenges. One potential issue is photobleaching, a phenomenon in which the absorber material degrades after prolonged exposure to high laser power. This degradation can reduce the efficiency of the signal conversion process and shorten the communication duration. In addition, if the light energy is increased too much, it can lead to optical breakdown of water, a process in which the water itself absorbs too much energy and breaks down, creating bubbles or cavitation that can interfere with signal transmission.

Despite these challenges, the photoacoustic air-to-underwater communication system offers several advantages over traditional communication methods. One of the key advantages is that it enables high-efficiency communication between air and underwater environments without the need for physical cables or other mechanical connections. This makes it ideal for applications such as underwater exploration, military communication, and environmental monitoring, where reliable and efficient data transmission is critical. The system's low energy requirements, combined with its cost-effective design, make it a promising solution for future developments in air-to-water communication technologies.

The photoacoustic air-to-underwater communication system represents a novel and highly efficient method for transmitting information across air and water interfaces. By utilizing the photoacoustic effect, the system can convert optical signals into acoustic waves, enabling effective communication in underwater environments. With its low energy requirements, cost-effective components, and ability to overcome the challenges of signal transmission in water, this system holds great potential for a wide range of applications. However, further research and development are needed to address the limitations, such as photobleaching and optical breakdown, to optimize the system for long-term use in challenging conditions.

Chapter 3

RESULT

The frequency of photoacoustic signals can be regulated by altering the laser, absorber, and detector, as deduced theoretically from the generation of photoacoustic waves, illustrated in Figures 3.1. Pure optical air-to-underwater communication struggles in complex water environments due to unstable refraction from waves and absorption in murky water. However, photoacoustic communication, which converts light into ultrasound, overcomes these issues. Experimental demonstrations of this concept are shown in Figure 3.1.

Figure 3.1 (a) presents images of the light path in still water, and with waves of 1 cm, 2 cm, and 3 cm, respectively. In each case, the laser spot on the bottom of the water tank shifts position (denoted as d), with a statistical analysis provided in Figure 3.1(b). Figure 3.1(c) explains that the position shift is due to the changing direction of the interface normal when a wave passes, which alters the refracted light's direction. While this minor shift of up to 2 cm at a 19 cm depth may seem small, in real-world conditions with larger waves and deeper water, the deviation could be tens of meters, leading to communication failure. Photoacoustic communication mitigates this issue by converting light into ultrasound, which is more tolerant of varying incident angles. Although vertical shifts in the waves can cause a time delay ($t = D/c$), the delay is negligible for high-frequency lasers, as demonstrated with a 1 kHz laser frequency where the measured time shift is much smaller than 1 ms. Figure 3.1(b) highlights the clear advantage of photoacoustic communication over laser communication under wavy conditions.

Polluted water presents significant challenges for underwater communication, primarily due to phenomena like plankton buildup on the surface, which increases light absorption. This absorption dramatically reduces the propagation distance of light signals,

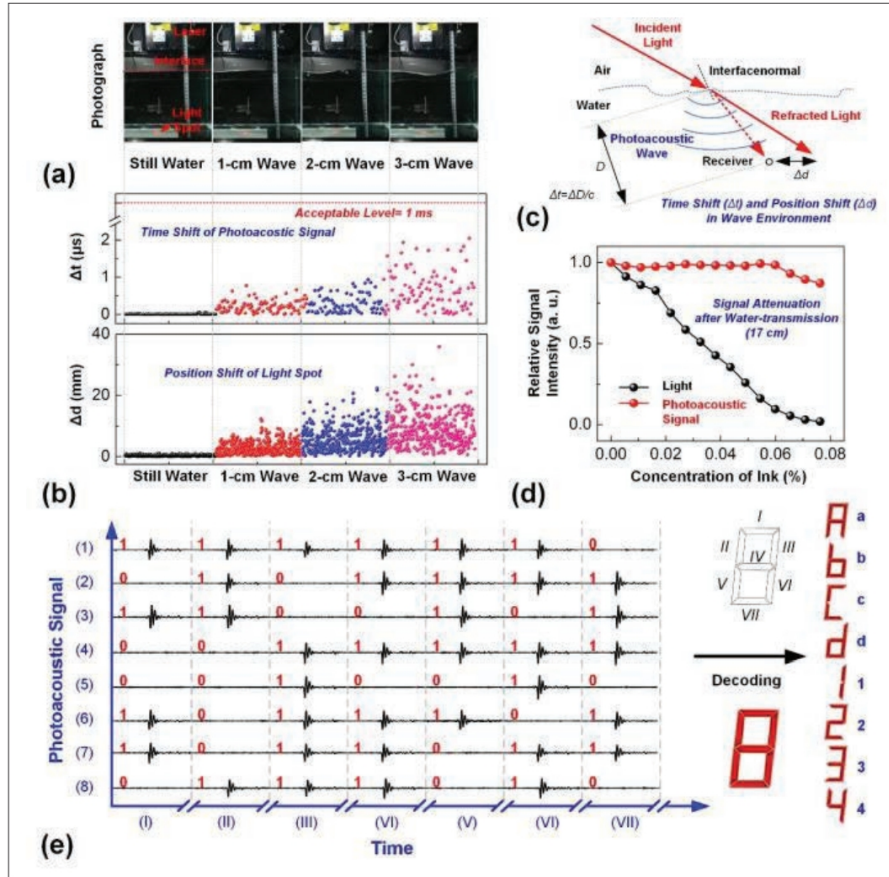


Figure 3.1: Communication demonstration in complex water environment

rendering traditional optical communication methods ineffective in such environments. However, the utilization of ultrasonic waves offers a promising alternative, as their longer wavelengths allow for effective transmission even in highly turbid waters.

To illustrate this capability, Figure 3.1(d) demonstrates the relative signal intensity of light and photoacoustic waves as they propagate through a black ink solution, simulating conditions found in polluted aquatic environments. The findings indicate a stark contrast: while light struggles to penetrate and is significantly attenuated, photoacoustic waves experience minimal energy loss, maintaining effective communication over a distance of 17 cm. This highlights the potential of photoacoustic communication systems to operate efficiently where traditional optical methods fail.

In a proof-of-concept experiment, a system for photoacoustic air-to-underwater communication was developed, focusing on the transmission of binary codes. The transmitted data stream was composed of alternating binary 1s and 0s. Figure 3.1(e) presents the photoacoustic signals captured in the time domain. The presence of a signal corresponds to

a binary 1, while the absence denotes a binary 0. By establishing a threshold line at half the peak amplitude of the signal, the binary codes can be distinctly identified, marking the initial phase of the decoding process.

For the demonstration, a seven-segment tube (SST) was employed for both coding and decoding the transmitted information. An example of this coding scheme is illustrated with the binary representation of the letter "a" as 1111110, where segments I through VI are illuminated while segment VII remains off. This system can be further developed and customized for various applications, adapting the encoding format to utilize other standards such as Morse code, ASCII, or even more sophisticated encrypted codes. The flexibility of this coding system allows for adjustments based on specific requirements, including the length of the communication and the level of security needed.

Overall, this research indicates a promising avenue for overcoming the limitations imposed by polluted water environments on communication technologies, paving the way for enhanced underwater interactions in diverse fields, including marine research, environmental monitoring, and underwater exploration.

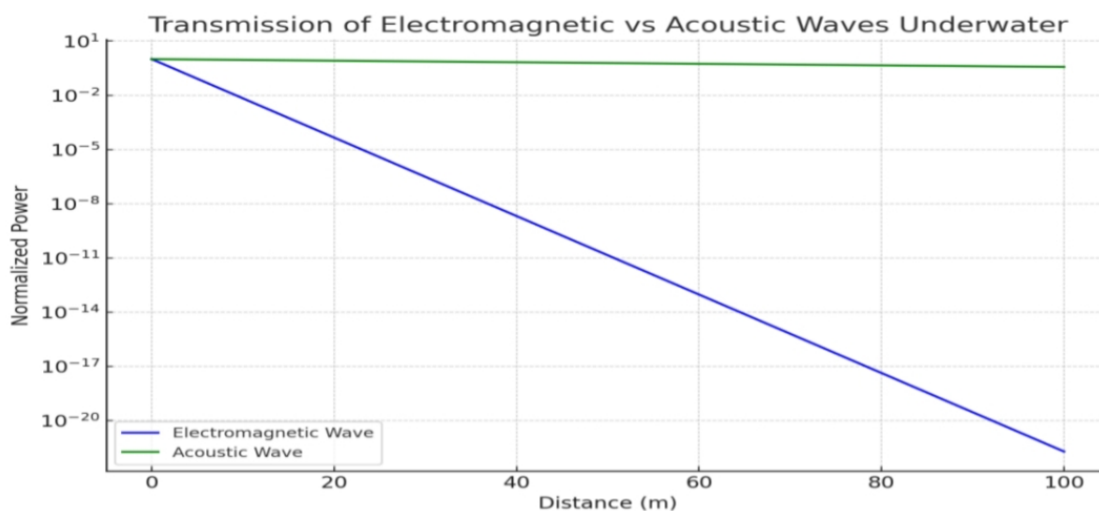


Figure 3.2: Comparison of Underwater Acoustic Wave and Electromagnetic Wave Attenuation

The figure 3.2 shows The comparison of electromagnetic and acoustic wave transmission underwater is crucial for understanding the limitations and potential of different communication methods in aquatic environments. This plot presents a clear demonstration of the attenuation characteristics of these two wave types as they travel underwater.

The X-axis represents the distance traveled, ranging from 0 to 100 meters, while the Y-axis, plotted on a logarithmic scale, indicates the normalized power of each wave type.

As shown in Figure 3.2, the descending blue line illustrates the behavior of electromagnetic waves as they travel through water. This line drops sharply with distance, indicating that these waves experience rapid power loss and cannot maintain effective signal strength over long distances underwater. This quick attenuation is primarily due to the high absorption of electromagnetic energy by water molecules, making these waves unsuitable for long-range underwater communication. Therefore, their application is largely limited to short-range, high-power scenarios.

In contrast, the green line represents acoustic waves, which remain nearly flat over the same distance, demonstrating minimal power loss. This characteristic allows acoustic waves to propagate much farther underwater with only slight attenuation, making them ideal for communication across substantial distances.

This comparison clearly illustrates why acoustic waves are predominantly used in underwater communication systems, while electromagnetic waves are restricted to short-range applications. The ability of sound to travel efficiently through water is essential for various uses, including submarine operations and underwater research.

Furthermore, the distinct propagation characteristics of these two types of waves influence the design of underwater communication systems. Acoustic systems can be optimized for long-range performance, taking advantage of the minimal signal loss. In contrast, electromagnetic waves may still be employed for immediate, short-range communications in specific situations, such as military operations or emergency responses.

Ultimately, the differences in how electromagnetic and acoustic waves propagate highlight the importance of selecting the right communication technology based on environmental conditions and distance requirements. Understanding these fundamental differences will be crucial in developing more effective underwater communication systems tailored to specific needs.

Chapter 4

FUTURE SCOPE

1. Improved Signal Clarity

Optimizing Signal Transmission: One of the major challenges in air-to-water communication is signal degradation during the transition between these mediums, which often leads to distortion and attenuation. Future advancements could:

- **Enhance relay design** to improve acoustic matching between air and water, reducing signal loss.
- Develop sophisticated **signal processing techniques** that filter out noise, providing clearer communication in noisy environments like busy coastal areas.
- Use **multi-frequency communication** to ensure that signals can remain strong and intelligible, even if some parts of the signal degrade during transmission.

2. Advanced Relay Materials

Innovations in Materials: The materials used in relays play a vital role in efficient signal transmission. Future breakthroughs could lead to more efficient relays capable of handling broader frequency ranges and signal strengths. This might include:

- **Piezoelectric materials** for converting acoustic signals into electrical signals and vice versa, enhancing relay sensitivity.
- **Metamaterials**, engineered to manipulate wave propagation, could minimize signal distortion and improve frequency response.

- **Nanotechnology** could help create more compact, durable relays that work effectively in challenging underwater environments.

3. Integration with the Internet of Things (IoT)

Enhancing Connectivity: Integrating passive relays with IoT networks would enable continuous communication between underwater devices and surface systems, paving the way for:

- **Real-time data transmission** from underwater sensors to surface platforms, facilitating better environmental monitoring, exploration, and maintenance.
- **Remote control and automation** of underwater systems, reducing the need for human intervention and allowing more autonomous operations.
- Leveraging **big data and AI** to analyze large datasets, detect patterns, and predict changes in underwater environments, such as temperature shifts or marine life activity.

4. Environmental Adaptability

Adjusting to Environmental Changes: The underwater environment is dynamic, with factors like salinity, temperature, and pressure fluctuating constantly. Future research could focus on creating relays that adjust to these variations to maintain reliable communication. This could involve:

- **Smart relays** equipped with sensors and AI to detect environmental changes and adapt signal transmission in real-time.
- **Self-regulating materials** that respond to environmental shifts, adjusting their physical properties to optimize signal propagation.
- Designing **robust relays** capable of withstanding extreme underwater conditions, such as deep-sea pressure or biofouling, to ensure consistent performance.

5. Cost Reduction

Lowering Costs: One of the key benefits of this technology is its potential for cost-effective deployment. Advances in manufacturing, materials, and signal processing could further reduce costs, leading to:

- **Broader adoption** of these systems across various industries, from small-scale environmental monitoring to large industrial operations.
- **Mass production** driving down unit costs, making large-scale deployments more practical and accessible.
- **Sustainable alternatives** to traditional methods like fiber-optic cables or sonar, offering more affordable, eco-friendly communication solutions.

6. Scalable Solutions

Expanding Network Deployment: As this technology evolves, deploying large-scale networks using passive relays will become feasible, leading to:

- **Ocean-wide sensor networks** for monitoring conditions like temperature, salinity, and pollutants across vast areas, which would be invaluable for climate studies and disaster prediction.
- **Industrial applications** in offshore sectors, such as oil and gas or renewable energy, where scalable relay networks could monitor underwater infrastructure and communicate with autonomous underwater vehicles.
- **Surveillance and defense** applications, with large relay networks providing extensive underwater coverage for detecting submarines, mines, or other threats, all with minimal human involvement.

Chapter 5

CONCLUSION

This paper introduces a photoacoustic air-to-underwater communication method based on low-cost passive relays. This novel approach is designed to facilitate communication across the air-water boundary, a historically challenging interface, using a combination of simulation and experimental testing. The proposed method achieves two key objectives: it significantly minimizes energy loss in both water and at the air-water interface and improves the ease of target acquisition in underwater environments. This makes the approach especially promising for applications in complex or dynamic water conditions, such as in regions with high sedimentation, turbidity, or biological activity.

One of the primary challenges in underwater communication is the loss of signal intensity due to attenuation in water, especially over long distances or in non-ideal conditions. Blue and green light, with wavelengths in the range of 450 nm to 550 nm, exhibit the lowest attenuation in pure water, making them the preferred choice for most underwater optical communication systems. This attenuation profile allows blue-green light to travel further in clear water compared to other visible light wavelengths, which experience higher absorption and scattering. However, using blue-green light for communication in complex water environments, such as those with high concentrations of organic particles or biological organisms, is challenging. Shorter wavelengths, including blue-green light, tend to scatter more readily when encountering particles such as plankton, leading to a rapid reduction in signal clarity and range. Furthermore, other visible light wavelengths encounter even greater attenuation and scattering, making them unsuitable for many underwater applications.

In addition to visible light, researchers have investigated other regions of the electro-

magnetic spectrum, including infrared, terahertz, and microwave frequencies. However, these wavelengths all suffer from significant drawbacks in underwater transmission. Infrared light has a high attenuation coefficient in water, resulting in rapid absorption and limited range. Similarly, terahertz waves, with their high frequencies, are almost entirely absorbed within a few centimeters in water, making them impractical for any significant underwater communication. Microwaves, though capable of short-range communication in very shallow waters, experience large attenuation as well and, therefore, are limited to surface or near-surface applications.

Ultrasonic waves, on the other hand, are well-suited for underwater transmission. These sound waves can travel relatively long distances in water with minimal attenuation, particularly within specific frequency ranges. Ultrasonic frequencies between 500 kHz and 2 MHz, for example, exhibit attenuation coefficients similar to those of blue-green visible light, making them an effective medium for underwater communication. Ultrasonic waves also tend to propagate more uniformly in water than light waves, making them resilient to certain environmental factors, like water turbidity and particle density. However, while ultrasonic waves are effective for long-range transmission, they do have limitations, including lower communication speeds compared to light-based methods.

The photoacoustic communication system leverages the advantages of ultrasonic waves while addressing some of their limitations. By utilizing the photoacoustic effect, which generates sound waves from pulsed laser light, this system produces communication signals in the megahertz range, which are ideal for underwater transmission. The conversion of laser energy into acoustic waves enables direct control over the frequency of the generated signals, allowing for optimized signal strength and transmission range. Furthermore, the system's reliance on passive relays reduces operational costs and maintenance, making it a viable option for extended deployments in challenging aquatic environments.

The complexity of target seeking is largely influenced by the directivity of the communication signals. In optical or light-based communication, the inherent directivity of light allows for focused, directional communication, minimizing interference and enhancing clarity. However, microwave and ultrasonic communication generally lack this level of directivity, often resulting in signals that scatter widely in water. Photoacoustic communication, as a hybrid technique, effectively combines the directional properties of laser light in air with the broad coverage of ultrasonic waves in water. This dual nature of

photoacoustic communication facilitates both precision in signal targeting and comprehensive area coverage, creating an ideal setup for applications where both are necessary. Nevertheless, this method is not without its challenges; for example, the speed of sound in water (1.5 km/s) is much slower than that of electromagnetic waves (225,000 km/s), which can result in slower data transfer rates.

Another technical hurdle in implementing this system is the alignment requirement between the airborne laser and the sea-based relay. This challenge is further amplified when the airborne source, such as an aircraft or drone, is in motion. Precise alignment is crucial to ensure that the laser targets the relay accurately, allowing for efficient signal conversion and transmission. To address this issue, the use of automated positioning systems and larger laser spots has been proposed. An automatic positioning system could allow for real-time adjustments to account for the movement of both the source and the relay, ensuring continuous alignment and communication. Additionally, employing a larger laser spot would reduce the need for exact alignment, as the expanded beam area would increase the likelihood of hitting the target.

The communication speed of this method can also be enhanced by using high-repetition-frequency lasers. This setup could increase the number of pulses transmitted per second, effectively boosting data transfer rates. However, the frequency of the ultrasonic waves acts as a limiting factor, as it dictates the maximum achievable data rate. Future developments could explore lasers with even higher repetition frequencies, provided they remain compatible with the frequency range of the generated ultrasonic signals.

Several advancements could further improve the performance of the photoacoustic communication system. For instance, establishing a bidirectional communication link would enable not only the transmission of data from air to underwater but also the retrieval of data from underwater sensors back to airborne receivers. Such a setup would expand the potential applications of this system to include real-time environmental monitoring, surveillance, and data collection, among others. Another potential enhancement involves increasing the laser energy, which could improve signal strength and the signal-to-noise ratio, extending the system's range. Using more efficient absorptive materials or contrast agents with cavitation effects could further increase the clarity of the signals, especially in murky or turbulent waters.

In conclusion, the photoacoustic communication method outlined in this paper presents

a promising solution for air-to-underwater communication. Through the integration of low-cost passive relays and a focus on efficient energy conversion, this system minimizes the common challenges associated with signal loss and target acquisition. By leveraging the directivity of lasers for targeting and the broadcast capabilities of ultrasonic waves in water, this approach effectively addresses the unique demands of air-to-underwater communication. While there are still areas for improvement—such as achieving higher data transfer speeds, refining alignment solutions, and developing bidirectional communication capabilities—the method holds significant potential. This technology could pave the way for advancements in marine research, environmental monitoring, and defense, providing a robust communication solution in even the most challenging aquatic conditions.

Bibliography

- [1] Z. Ji, Y. Fu, J. Li, Z. Zhao and W. Mai, "Photoacoustic Communication from the Air to Underwater Based on Low-Cost Passive Relays," in IEEE Communications Magazine, vol. 59, no. 1, pp. 140-143, January 2021, doi: 10.1109/MCOM.001.2000607.
- [2] Mahmud M, Islam MS, Ahmed A, Younis M, Choa FS. Cross-Medium Photoacoustic Communications: Challenges, and State of the Art. Sensors (Basel). 2022 Jun 1;22(11)
- [3] Md Shafiqul Islam, Mohamed Younis, Muntasir Mahmud, Gary Carter, Fow-Sen Choa, A peak detection based OOK photoacoustic modulation scheme for air to underwater communication, Optics Communications, Volume 529, 2023, 129078, ISSN 0030-4018,
- [4] X. Wang et al., "Microwave-Induced Thermoacoustic Communications", IEEE Trans. Microwave Theory Tech., vol. 65, no.9, 2017, pp.3369-78
- [5] D. He, "High-Power Nd: YAG-Generated Underwater Sound Source for Air-Submarine Communication," Solid State Lasers VIII, Int'l. Society for Optics and Photonics, vol. 3613, 1999, pp. 83–92.
- [6] H. Chen, Y. Shi, and D. Xing, "Photoacoustic Thermorelaxation Microscopy for Thermal Diffusivity Measurement," Opt. Lett., vol. 44, no. 13, 2019, pp. 3366–69.
- [7] F. Blackmon, L. Estes, and G. Fain, "Linear Optoacoustic Underwater Communication," Appl. Opt., vol. 44, no. 18, 2005, pp. 3833–45.

APPENDIX

- Reference Paper of the Seminar

Photoacoustic Communication from the Air to Underwater Based on Low-Cost Passive Relays

Zhong Ji, Yong Fu, Jinliang Li, Zhijuan Zhao, and Wenjie Mai

With the development of ocean research, communications from the air to underwater (e.g., aircraft to submarines) are especially important. However, the existing communication methods have their own limitations, and wireless communication between underwater and the air has remained an unsolved problem.

ABSTRACT

With the development of ocean research, communications from the air to underwater (e.g., aircraft to submarines) are especially important. However, the existing communication methods have their own limitations and wireless communication between underwater and the air has remained an unsolved problem. In this article, we propose a photoacoustic communication based on low-cost passive relays, which is capable of not only minimizing the energy loss both in water and in the air-water interface, but also relieving the difficulty of target seeking. Thus, we can get a six order of magnitude reduction of laser energy (25 J vs. 27 μ J) using passive relays whose cost is less than US\$1.00. It is also demonstrated that our photoacoustic communication can work decently under complex water situations including waving and pollution. This creationary photoacoustic application may become a practical way to communicate from the air to underwater.

INTRODUCTION

For decades, wireless communication between underwater and the air has remained an unsolved problem. Communications between the air and underwater are increasingly important with the development of marine technology and the exploration of the underwater world. Recently, one-way communication from underwater to the air was realized by an MIT group in 2018 with the development of a translational acoustic-radio frequency communication technology [1]. However, the other way communication, from the air to underwater, using the existing communication technologies still has its own limitations in practical applications. Among them, direct communication is very hard. For example, light communication works well in air or underwater [2]; however, the laser beam is reflected and refracted at the air-underwater interface, which leads to large energy loss and low targeting stability in the presence of waves. Although the attenuation coefficient of blue/green light is small in pure water [3], it is difficult to adapt to complex water environments (e.g., plankton) due to high scattering of short wavelength. The attenuation coefficient of electromagnetic wave with other wavelengths is too large, leading to high energy loss [4, 5]. Specifical-

ly, microwave or radio wave, as the gold medium for air communications, cannot travel in water [5]. For acoustic communication, although ultrasonic attenuation in air and water is small, the acoustic impedance difference of air-underwater interface is very large [6], so ultrasonic waves lose most of the energy when passing through the interface.

Active relays are widely used in conventional air-underwater communication [7, 8]. The active relays need to be powered so that they can actively convert incoming signals (e.g., radio or lightwaves) from the air into underwater signals (e.g., ultrasonic or lightwaves), just like a traditional radio receiver, which repeats the signal as sound waves below the surface. These devices are usually installed in fixed offshore base stations, buoys, or ships. However, active relays have some defects leading to limited applications. First, active relays themselves are expensive and need regular maintenance, resulting in relatively high costs. Second, it is hard to achieve full coverage of active relays in an open ocean, which may lead to communication failure. Finally, active relays have relatively large volume, resulting in some concerns about communication security. Therefore, passive relays in air-to-underwater communication are urgently needed.

Converting electromagnetic waves into ultrasonic waves at the air-underwater interface for communication is a potential solution, because electromagnetic waves can transmit in the air and ultrasonic waves underwater with low energy loss. The most common mechanisms of converting electromagnetic waves into ultrasonic waves are thermal expansion, surface evaporation, and optical breakdown [9]. That is, water in the air-water interface is a possible passive relay. Several research groups have made some exciting achievements [9–12]. Among them, Wang *et al.* used pulsed microwave to generate pulsed ultrasonic waves by a thermal expansion mechanism because the absorption coefficient of water is relatively large in the microwave band [10]. But microwaves are diverging, so a high-power microwave source is required in order to deliver enough energy to the relay. In the visible light and near infrared band, other researchers also have to adopt high-power lasers to produce ultrasonic waves [9, 11, 12] due to low absorption coefficients. Thus, they have to adopt high-power lasers

with single pulse energy of tens of joules [11], which severely handicaps the application.

The photoacoustic effect, also called the optoacoustic effect, in the usual sense is a more efficient way to convert light energy into ultrasound energy at light absorbers by thermal expansion; when the absorber is illuminated with a pulsed laser, a small temperature rise can be produced to generate a photoacoustic wave under the conditions of thermal and stress confinement. These merits have resulted in the photoacoustic effect being widely applied in the biomedical field [13] and industrial applications [14]. Meanwhile, contrast agents with high optical absorption coefficient can improve the efficiency of energy conversion because the intensity of photoacoustic signal is proportional to the absorption coefficient. All of the above bring possibilities for high-efficiency photoacoustic air-to-underwater wireless communication.

In this article, we introduce some low-cost and high-efficiency passive relays in the proposed photoacoustic air-to-underwater communication. We expect the proposed communication method with passive relays can overcome the high energy loss of pure electromagnetic waves in water and high reflection of pure ultrasonic waves in the air-underwater interface. Moreover, the introduction of passive relays with high optical absorption coefficient can greatly improve communication efficiency, and thus get a six-order-of-magnitude reduction of laser energy (25 J [11] vs. 27 μ J) using passive relays whose cost is less than US\$1.00. Through simulations and experiments, it can be concluded that this novel photoacoustic application may become a practical way to communicate from the air to underwater.

METHODS AND EXPERIMENT SECTION

The schematic diagram of photoacoustic air-to-underwater communication is displayed in Fig. 1. The sender in the air (e.g., aircraft) uses pulsed laser to emit a coded laser beam (e.g., 11011) with information. After transmission in the air, the laser radiates to the signal relay, which is a light absorbing material released by the receiver (e.g., submarine). Pulsed ultrasonic signals (i.e., photoacoustic waves) with the same code as the laser are generated from the signal relay due to the photoacoustic effect. After transmission in water, the photoacoustic waves can be detected by the receiver; then communication is finished by decoding. When an absorber is illuminated by a pulse laser, a small transient temperature rise can be produced; then the heated absorber thermally expands and contracts, and finally the photoacoustic wave comes out. To demonstrate photoacoustic air-to-underwater communication, we adopted a simply equipped experimental setup, which is similar to the single-element photoacoustic imaging system or ultrasonic-based electromagnetic wave detection system.

In this article, as a message sender, the laser (custom-made, Beijing Laize, China) triggered by the synchronization pulse of the function generator (AFG2021, Tektronix, U.S.) delivers a laser beam with wavelength of 660 nm and 14 ns pulse width at a repetition frequency of 1 kHz. The function generator and laser act as the information sender. The single pulse energy of the laser

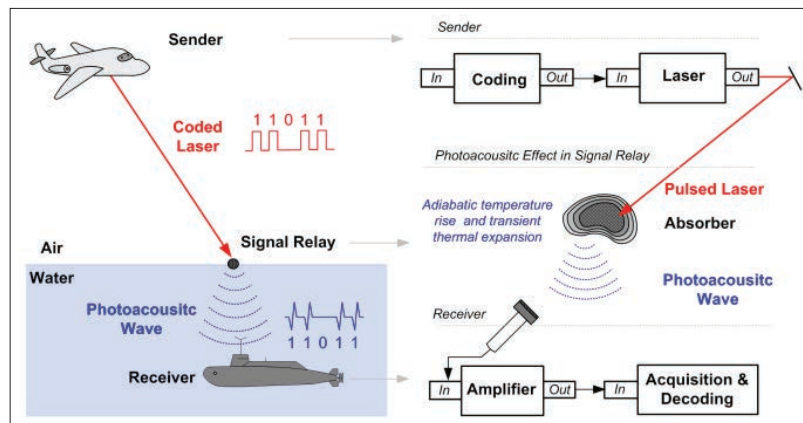


FIGURE 1. Design of photoacoustic air-to-underwater wireless communication based on low-cost passive relays.

is 27 μ J, which is far less than tens of joules for breaking down water. The ultrasonic signal receiver is composed of an ultrasonic detector (5 MHz: I5P6NF, Doppler, China; or 500 kHz: DYW-500-E, Dayu Electric., China), a home-made amplifier with a magnification of about 40 dB [15], an oscilloscope (MDO3054, Tektronix, U.S.), and a computer. The breakdown of the cost and volume of our system is estimated and provided in the supplementary materials. When the laser beam reaches the signal relay, ultrasonic waves are generated by the photoacoustic effect. After underwater transmission for a distance, the generated waves, namely photoacoustic waves, can be received by an ultrasonic detector. Then the signal can be amplified by the home-made amplifier, and then be acquired by the oscilloscope and decoded by the computer. Here, the ultrasonic detector, amplifier, oscilloscope, and computer are acting as the information receiver. Therefore, the photoacoustic air-to-underwater communication system is established.

In the above conditions, we can obtain photoacoustic signals whose voltage ranges from ~100 mV to 1 V. To the best of our knowledge, there are no reports about photoacoustic communication from the air to underwater based on low-power light source and low-cost passive relays. Because the photoacoustic signal increases linearly with light energy, theoretically we can use a laser with dozens of millijoules to realize the effect of existing kilowatt-level sonar. It should be noted that under such a strong laser, the absorbers may be photobleached, leading to short communication duration. If the light energy continues to increase to dozens of Joules, optical breakdown of water may occur, which is another topic. In conclusion, we have got a six-order-of-magnitude reduction of laser energy using passive relays whose cost is less than US\$1.00.

RESULTS

We can regulate the frequency of photoacoustic signals by changing the laser, absorber, and detector by theoretical deduction of the generation of photoacoustic waves, as shown in Figs. 1 and 2. Pure light air-to-underwater communication cannot work well in a complex water environment mainly because of unstable refraction directions in waves as well as absorption of muddy water. Photoacoustic communication can overcome these problems by converting light into

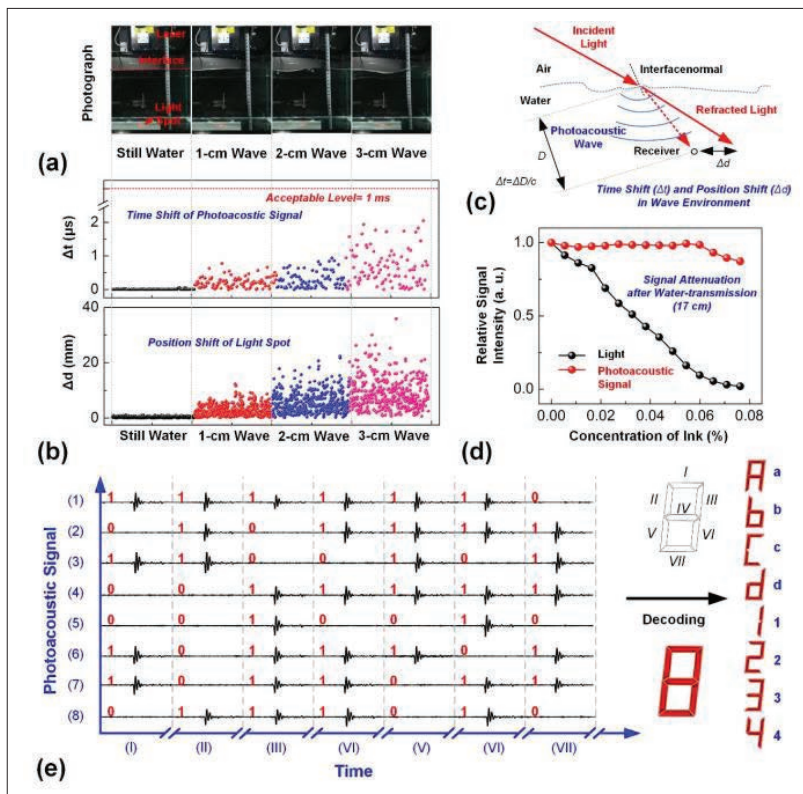


FIGURE 2. Communication demonstration in complex water environment: a) photographs of communication path in still water, 1 cm wave, 2 cm wave, and 3 cm wave, respectively; b) measured time shift of photoacoustic signal (Δt) and position shift of light spot (Δd) in different magnitude of waves; c) signal propagation path; d) relative signal intensity of light and photoacoustic waves after propagation in black ink solution to simulate communication in polluted water environment; e) raw time-domain signals and decoding process to demonstrate photoacoustic communication.

ultrasound, and experiments are performed to demonstrate the capabilities, as shown in Fig. 2. Figure 2a shows photographs of light path in still water, 1 cm wave, 2 cm wave, and 3 cm wave, respectively. The laser spot on the bottom of the water tank has position shift (Δd) in each photograph, and accurate statistical analysis is shown in Fig. 2b. Figure 2c explains why position shift is generated. When the wave comes, the direction of interface normal changes accordingly; then the direction of the refracted light changes. The direction change would be lethal to pure light air-to-underwater communication, because one small direction change makes a great difference in light spot. That is, although there is only maximal 2 cm shift when receiving depth is 19 cm in this experiment, the random deviation will be tens of meters in actual situations (with large waves and in deep water), causing communication failure. However, photoacoustic communication can avoid this problem by converting light into ultrasound, which can accept a large incident angle. Indeed, waves can also cause a distance shift (ΔD) in the vertical direction, leading to a signal time shift ($\Delta t = \Delta D/c$). In consideration of 1 kHz laser frequency, the measured time shift can be ignored ($2 \mu s \ll 1$ ms). Figure 2b clearly demonstrates the advantage of our photoacoustic communication compared to the laser communication in wave condition.

Another possible factor that can affect communication is polluted water, like plankton gathered on the water surface. In this condition,

light absorption increases dramatically, leading to extremely short light propagation length, but photoacoustic wave (ultrasonic wave) with longer wavelength will greatly alleviate the problem. Figure 2d shows the relative signal intensity of light and photoacoustic wave after propagation in black ink solution to simulate communication in polluted water environment. The results indicate that light can hardly propagate in this environment, but photoacoustic waves can arrive with very low energy loss in 17 cm distance. Overall, we proved experimentally that pure light communication can hardly work in air-to-underwater condition in complex water environments, but photoacoustic communication can work well.

Experiments are performed to demonstrate proof-of-concept photoacoustic air-underwater communication. The information to be communicated is binary codes, a data stream containing several binary 1s as well as several binary 0s. As shown in Fig. 2e, obtained photoacoustic signals in the time domain have alternation of presence and absence depending on code 1 or 0. Drawing a line at half peak of the signal can identify code 1 or 0, which is the first step of the decoding process. Next, these codes are translated according to the prearranged code book. As a demonstration, seven segment tube (SST) was adopted for coding and decoding. To be specific, brightness or darkness of the individual segment tube (I, II, III, IV, V, VI or VII) depends on the corresponding bit of codes. For example, code 1111110 means brightness of I, II, III, IV, V, and VI, and darkness of VII, so the code can be decoded as letter a. It should be noted that the codes are dependent on length and security of communication information, so Morse code, American standard code for information interchange (ASCII) code, or other complex encrypted code can be used in future applications.

DISCUSSION

Generally speaking, blue and green light (450 nm to 550 nm) has the minimum attenuation coefficient in pure water [4, 5], so these bands are widely used in underwater optical communication. However, it is difficult to adapt to a complex water environment due to high scattering of short wavelength (e.g., plankton in the water). Visible light with other wavelengths has a larger attenuation coefficient and a similar high scattering problem. Infrared light, terahertz wave, and microwave [5] have large attenuation coefficients, so they are basically untransmissible in water. As for ultrasonic waves [6], water can be regarded as a low pass filter, and an ultrasonic wave with about 500 kHz to 2 MHz has a close attenuation coefficient with visible light. Considering the original photoacoustic signal spectra and communication speed, photoacoustic signals with megahertz are suitable for communication.

Target-seeking complexity depends directly on the directivity of the information carriers. In comparison to pure light communication (good directivity), microwave communication (poor directivity) and ultrasonic communication (poor directivity), photoacoustic communication, as a hybrid method of communication, inherits good directivity of laser in the air and broadcast transmission of ultrasonic waves underwater. Certainly, photoacoustic communication has some disadvantages; for example,

low communication speed due to low sound velocity in water (~1.5 km/s) in contrast with electromagnetic wave (~225,000 km/s).

There are still some challenges for the proposed communication method, for example, the misalignment issue between the following-sea relay and fast-moving aircraft. One possible way to solve that is using an automatic positioning system and adopting relatively large light spot. Another challenge is relatively low communication speed. The speed of the proposed communication method can be improved by using high-repetition-frequency laser, but the upper limit is determined by ultrasonic frequency. Moreover, bidirectional communication links would be more helpful in this air-underwater scenario [1]. Finally, increasing the laser energy appropriately and introducing absorbers with higher efficiency, or contrast agents with cavitation effect, are beneficial to increase signal-to-noise ratio and thus increase communication distance.

CONCLUSIONS

In conclusion, we propose a photoacoustic air-to-underwater communication method based on low-cost passive relays in this article. Through numerical simulation and experimental verification, it is concluded that the proposed communication is capable of not only minimizing the energy loss both in water and in the air-water interface, but also relieving the difficulty of target seeking. Our method is a promising way to communicate from the air to underwater, especially in complex water conditions.

ACKNOWLEDGMENTS

Authors acknowledge the financial supports from the National Natural Science Foundation of China (Grant Nos. 61604061, 52002147, 51772135 and 61875074), the Natural Science Foundation of Guangdong Province, China (Grant Nos. 2017A020215135 and 2018A030310659), the Science and Technology Program of Guangzhou, China (Grant No. 201804010432), the Young Innovative Talent Project of General Colleges and Universities in Guangdong Province (No. 2018KQNCX007), the Fundamental Research Funds for the Central Universities (Grant No. 21618311) and the China Postdoctoral Science Foundation (Grant No. 2019M663363).

REFERENCES

- [1] F. Tonolini and F. Adib, "In Networking across Boundaries: Enabling Wireless Communication through the Water-Air Interface," *Proc. 2018 Conf. ACM Special Interest Group on Data Commun.*, 2018, pp. 117–31.
- [2] S. Gaiaarin et al., "Dual-Polarization Nonlinear Fourier Transform-Based Optical Communication System," *Optica*, vol. 5, no. 3, 2018, pp. 263–70.
- [3] F. M. Sogandares and E. S. Fry, "Absorption Spectrum (340–640 nm) of Pure Water. I. Photothermal Measurements," *Appl. Opt.*, vol. 36, no. 33, 1997, pp. 8699–8709.

- [4] C. Patel and A. Tam, "Optical Absorption Coefficients of Water," *Nature*, vol. 280, no. 5720, 1979, p. 302.
- [5] G. W. King, R. Hainer, and P. C. Cross, "Expected Microwave Absorption Coefficients of Water and Related Molecules," *Phys. Rev.*, vol. 71, no. 7, 1947, pp. 433–43.
- [6] O. A. Godin, "Transmission of Low-Frequency Sound through the Water-to-Air Interface," *Acoust. Phys.*, vol. 53, no. 3, 2007, pp. 305–12.
- [7] N. Celik et al., "A Software-Defined Opto-Acoustic Network Architecture for Internet of Underwater Things," *IEEE Commun. Mag.*, vol. 58, no. 4, 2020, pp. 88–94.
- [8] N. S. Gupta et al., "Asymmetric Satellite-Underwater Visible Light Communication System for Oceanic Monitoring," *IEEE Access*, vol. 7, 2019, pp. 133,342–50.
- [9] S. Zong et al., "Study of Opto-Acoustic Communication Between Air and Underwater Carrier," *4th Seminar on Novel Optoelectronic Detection Technology and Application*, Int'l. Society for Optics and Photonics, vol. 10697, 2018, p. 106975S.
- [10] X. Wang et al., "Microwave-Induced Thermoacoustic Communications," *IEEE Trans. Microwave Theory Tech.*, vol. 65, no. 9, 2017, pp. 3369–78.
- [11] F. Blackmon, L. Estes, and G. Fain, "Linear Optoacoustic Underwater Communication," *Appl. Opt.*, vol. 44, no. 18, 2005, pp. 3833–45.
- [12] D. He, "High-Power Nd: YAG-Generated Underwater Sound Source for Air-Submarine Communication," *Solid State Lasers VIII*, Int'l. Society for Optics and Photonics, vol. 3613, 1999, pp. 83–92.
- [13] A. De La Zerda et al., "Carbon Nanotubes as Photoacoustic Molecular Imaging Agents in Living Mice," *Nature Nanotech.*, vol. 3, no. 9, 2008, pp. 557–62.
- [14] H. Chen, Y. Shi, and D. Xing, "Photoacoustic Thermorelaxation Microscopy for Thermal Diffusivity Measurement," *Opt. Lett.*, vol. 44, no. 13, 2019, pp. 3366–69.
- [15] Y. Fu et al., "Thermoacoustic Imaging over Large Field of View for Three-Dimensional Breast Tumor Localization: A Phantom Study," *Med. Phys.*, vol. 41, no. 11, 2014, pp. 110701.

BIOGRAPHIES

ZHONG JI (jizhong@jnu.edu.cn) received his B.S. degree in electronic science and technology (2010) from Tianjin University and his Ph.D. degree in optics (2016) from South China Normal University (SCNU). He is now an assistant professor at Jinan University (JNU). Since 2010, his main research interest is photoacoustic/thermoacoustic imaging.

YONG FU (fuyong@jnu.edu.cn) received his B.S. degree in electronic information engineering (2011) from Nanhua University and his Master's degree in optics (2014) from SCNU. He is now a laboratory technician at JNU. His main research interest is photoacoustic/thermoacoustic imaging.

JINLIANG LI (lijinliang@email.jnu.edu.cn) received his B.S. degree in electronic science and technology (2012) from South China University of Technology (SCUT) and his Ph.D. degree in physics (2017) from East China Normal University. He is now an associate professor at JNU. His main research interest is in Na/K-ion batteries.

ZHIJUAN ZHAO (zhijuanzhao@email.jnu.edu.cn) received her B.S. degree in electronic science and technology (2010) from Henan University and her Ph.D. degree in physics (2017) from Xiamen University. She is now an associate professor at JNU. Her main research interest is in electrochromic devices.

WENJIE MAI (wenjiemai@email.jnu.edu.cn) received his B.S. degree in physics (2002) from Peking University and his Ph.D. degree in materials science and engineering (2009) from Georgia Institute of Technology. He is now a professor at JNU. His main research interests include energy conversion, harvesting, and storage devices, such as solar cells, nanogenerators, supercapacitors, batteries, and photoelectrochemical water splitting.

Through numerical simulation and experimental verification, it is concluded that the proposed communication is capable of not only minimizing the energy loss both in water and in air-water interface, but also relieving the difficulty of target-seeking. Our method is a promising way to communicate from the air to underwater, especially in complex water conditions.



Department of Electronics & Communication
Vidya Academy of Science & Technology
Thalakkottukara, Thrissur - 680 501
(<http://www.vidyaacademy.ac.in>)